

Convolution + Pooling Exercises (Questions and Answers)

Instructions: Use valid convolution (no padding). After convolution, apply 2×2 pooling on the convolution output. Stride values are given per question.

Part A: Questions

Q1.

Convolution: 3×3 kernel, stride 1.

Pooling: 2×2 max pooling, stride 2.

Image (6×6):

1	2	3	4	5	6
2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11

Kernel (3×3):

-1	0	1
-2	0	2
-1	0	1

Q2.

Convolution: 3×3 kernel, stride 2.

Pooling: 2×2 average pooling, stride 1.

Image (6×6):

5	1	2	0	3	4
4	2	1	3	2	0
0	1	5	2	4	1
2	3	0	1	2	5
1	4	2	3	0	2
3	0	1	4	1	2

Kernel (3×3):

1	1	1
1	1	1
1	1	1

Q3.

Convolution: 3×3 kernel, stride 1.

Pooling: 2×2 average pooling, stride 2.

Image (6×6):

1	0	1	0	1	0
0	1	0	1	0	1
1	0	1	0	1	0
0	1	0	1	0	1
1	0	1	0	1	0
0	1	0	1	0	1

Kernel (3×3):

0	1	0
1	-4	1
0	1	0

Q4.

Convolution: 3×3 kernel, stride 1.

Pooling: 2×2 max pooling, stride 1.

Image (6×6):

10	10	10	10	10	10
10	20	20	20	20	10
10	20	50	50	20	10
10	20	50	50	20	10
10	20	20	20	20	10
10	10	10	10	10	10

Kernel (3×3):

-1	-2	-1
0	0	0
1	2	1

Q5.

Convolution: 3×3 kernel, stride 2.

Pooling: 2×2 max pooling, stride 1.

Image (6×6):

2	2	2	2	2	2
2	4	4	4	4	2
2	4	8	8	4	2

2	4	8	8	4	2
2	4	4	4	4	2
2	2	2	2	2	2

Kernel (3×3):

0.11	0.11	0.11
0.11	0.11	0.11
0.11	0.11	0.11

Part B: Answers (Final Outputs Only)

Q1 Answer

Convolution output:

8	8	8	8
8	8	8	8
8	8	8	8
8	8	8	8

Pooling output:

8	8
8	8

Q2 Answer

Convolution output:

21	22
18	19

Pooling output:

20

Q3 Answer

Convolution output:

-4	4	-4	4
4	-4	4	-4
-4	4	-4	4
4	-4	4	-4

Pooling output:

0	0
0	0

Q4 Answer

Convolution output:

60	130	130	60
30	90	90	30
-30	-90	-90	-30
-60	-130	-130	-60

Pooling output:

130	130	130
90	90	90
-30	-90	-30

Q5 Answer

Convolution output:

3.33	4.22
4.22	5.78

Pooling output:

5.78
